# Long-Lead Critical Components Facility-Class Design Basis

The First Facility Class of the Critical Manufacturing Sector Design Basis

DBA-CM-LL-FC1 — Facility-Class Design Basis, DBA-CM Series

Issued Under: Critical Manufacturing Sector Framework (DBA-CM_Sector_Framework v0.3); DBA-MA Parent Framework

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# 1. Purpose and Scope

## 1.1 Purpose

This document sets out the design basis for long-lead critical components, the first facility class in the DBA-CM series. The *Critical Manufacturing Sector Framework* (DBA-CM_Sector_Framework) states the sector's design basis in general terms, and this document applies it to one kind of facility: the plant that builds the large, custom, slow-to-make components on which other sectors' recovery depends. It names the events such a plant and the nation's capacity of such plants must withstand, traces how each event reaches the facilities that are waiting for what the plants make, sets the parameters that size the class's protection, and states what must be true of the capacity, the order book and the reserve for a damaged facility elsewhere to be restored in a bounded time. It describes no particular plant. A design basis for a specific plant, manufacturer or component starts from this document and adds only what is particular to it.

The framework calls Critical Manufacturing the upstream-producer sector, whose consequence is almost entirely downstream: a plant's loss is rarely a large event on site, and its significance is that what it makes cannot be replaced, so that the sector that depends on it cannot recover (framework §1.5). It adopts RT_HVLLC, the recovery timeline for high-value long-lead components, as its native governing metric, applied not to the damaged asset but to the manufacturing capacity that bounds its replacement (framework §3.6), and it names this class the flagship of the sector and its Phase-A instrument (framework §3.2, §9). This document takes both up. On the author's decisions of 14 October 2026 the class is the sector's first, ahead of Precision and Electronics (DBA-CM-PE-FC3), and proposes three reference events of its own (Section 9.2).

## 1.2 Scope

The class covers the facilities that make the components the framework's register names: large power transformers, large generators and turbines, high- and extra-high-voltage equipment, and the large castings and forgings that underlie them (framework §3.2, §8.2). For a plant it covers the design and engineering of a custom unit; the receipt of the materials it is made from, above all the grain-oriented electrical steel of a transformer's core and the heavy forgings of a turbine's rotor; the manufacturing process, from core stacking and winding to vapor-phase drying and tanking, or from forging to machining and balancing; the high-voltage test bay or the test stand in which a unit is proven; and the dispatch of a unit that may weigh several hundred tons. It includes the plant's order book, because the time a damaged facility waits for its replacement is set as much by the queue in front of it as by the time to build. And it includes the people who design, build and test such units, who are few, skilled and slow to train.

Three neighbors are separate classes of this series. The specialty steels, alloys and large forgings that feed this class, and the processing of strategic materials, are *Primary Metals and Strategic*

*Materials* (DBA-CM-PM-FC2). The semiconductors and control-system components that instrument the units this class makes are *Precision and Electronics Manufacturing* (DBA-CM-PE-FC3). Supply-chain risk management, the Defense Production Act's Title III mobilization and the strategic-spares reserve are the coordination class, DBA-CM-CorpCo. They are covered here only where the plant depends on them. Where a plant forges its own rotors, the forge is this class's; where it buys them, the forge that makes them is the primary-metals class's, and its loss is a design-basis event here (Section 5.2).

Two boundaries with other series are drawn in the framework and taken here as given (framework §6.1, §6.2). The installed asset, the transformer in its substation or the turbine in its plant, is the energy series', which states its RT_HVLLC for each facility class (*The RT_HVLLC Gap*, the energy series' strategic-reserve paper, §2). What this class states is the manufacturing capacity that bounds that figure. And the heavy-haul movement of an oversized unit from plant to site is shared with the transportation series' conveyance class (*Conveyances and Fleet*, DBA-TRAN-CV-FC3), the framework naming the large power transformer as itself a boundary object (framework §6.2). Section 8.3 states the boundaries.

What makes a facility a member of this class is the same at every site: it makes, to order and slowly, a component that another sector cannot run without and cannot buy off a shelf. The particular component, process and order book differ from plant to plant, and that detail belongs in each plant's own design basis.

## 1.3 The Reference Facilities

Three kinds of plant are used as reference cases. Each was chosen because it shows a different way the capacity to replace a long-lead component can be lost.

The large-power-transformer plant designs, builds and tests transformers and autotransformers for transmission substations and generating stations. It is the governing configuration for the class, because the large power transformer is the component the energy series names first among those with no ready spare, with an RT_HVLLC of eighteen to twenty-four months under normal conditions (*The RT_HVLLC Gap* §2.1), and because the nation depends on few such plants: the National Infrastructure Advisory Council's report to the President of June 2024, published in a version marked draft and pre-decisional, put domestic production at only about twenty percent of large-power-transformer demand (National Infrastructure Advisory Council, *Addressing the Critical Shortage of Power Transformers to Ensure Reliability of the U.S. Grid*, June 2024). The closure of Ukraine's largest transformer plant by the war in 2022, part of the event conferred by the framework as DBA-CM-RE-1 (framework §9.1; Section 6.3), is the case; the plant's status after March 2022 is open, the government having taken control of the company under martial law in November 2022 (*The Denial of Large-Transformer Replacement in Ukraine* §2.2).

The large-turbine-and-generator plant builds steam and combustion turbines, generators and their rotors. It bounds the class's longest timelines: the energy series puts the RT_HVLLC of a supercritical steam turbine rotor at twenty-two to thirty months (*The RT_HVLLC Gap* §2.3). Its

output depends on heavy forgings that only a few presses in the world can make, which ties it to the primary-metals class at the point where a single upstream loss reaches every unit downstream.

The heavy-castings-and-forgings plant pours and forges the large steel parts on which the other two depend. It is shared with the primary-metals class and is used here as the reference for the plant whose production equipment is itself large, hazardous and controlled by networked systems, so that an attack on its controls damages the capacity directly. The attack on a German steel mill reported by the Federal Office for Information Security in December 2014, conferred by the framework as DBA-CM-RE-3 (framework §9.1; Section 5.5), is the case.

## 1.4 Class Type

This is a physical facility class under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical). The harm it governs runs through plants, their heavy equipment and their test bays, and through the people who run them. In the framework's structure it sits under Long-Lead Critical Components (LL), the first of the sector's three function classes (framework §3.1, §3.2). It is unlike the physical classes before it in one respect the framework states plainly: its consequence is not on site. A transformer plant that burns harms few people where it stands, and its design basis is written for the facilities, elsewhere and in other sectors, that it would have supplied.

## 1.5 Relationship to Governing Documents

This instrument is issued under the Critical Manufacturing Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, bounding principles, consequence measure or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. Section references of the form framework §N resolve against the Critical Manufacturing Sector Framework in force. RT_HVLLC is the energy series' metric, established there and in the strategic-reserve paper, *The RT_HVLLC Gap*, both Project 8's; this class cites their values and does not restate their derivation.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term envelope. Placement is on the consequence of the individual action, assessed per action; this

instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

# 2. The Long-Lead Critical Components Facility Class

## 2.1 Defining Characteristics

Five characteristics define the class, and each is a reason the loss of a plant reaches farther and lasts longer than the loss of most facilities.

The first is the custom unit. A large power transformer or a turbine rotor is designed for its site: its voltage, impedance, rating and physical dimensions are set by the substation or the plant it will serve. A unit built for one site may not fit another, so that a spare must be specified in advance and a damaged unit cannot simply be swapped for whatever is available.

The second is the long build. A unit takes months to design, build, dry, test and ship, and the steps cannot be compressed much by effort. The energy series' RT_HVLLC values, eight months for a combustion-turbine hot section to thirty for a supercritical rotor, are, in that paper's account, the lead times manufacturers have reported under normal supply-chain conditions (*The RT_HVLLC Gap*, Executive Summary).

The third is the queue. A damaged facility does not wait only for its unit to be built; it waits for every unit ahead of it in the order book. Under normal conditions the queue is part of the reported lead time. When demand rises, it lengthens, and the RT_HVLLC of every facility that depends on the plant lengthens with it, without anything having been damaged.

The fourth is concentration. For each component a few plants in the world can build the largest units, many of them abroad, and they depend on a few sources of their critical materials. The framework calls this the sector's defining exposure (framework §2.1, §6.3).

The fifth is the heavy unit. A large power transformer can weigh several hundred tons. It moves by rail on special cars or by road on multi-axle trailers, often by sea, and sometimes cannot move at all where bridges or routes have been lost. The time to deliver is part of the time to recover.

## 2.2 The Substitutability and Lead-Time Sub-Index

The framework sub-indexes the sector's classes by substitutability and manufacturing lead time, so that the design-basis load a plant carries scales with how irreplaceable its output is (framework §3.1, §8.2). This instrument states three tiers for the class.

The first tier is the plant that is one of very few able to build a component whose RT_HVLLC exceeds a year and for which no strategic spare exists in the specification it serves: the plants that build extra-high-voltage autotransformers, converter transformers and the largest turbine rotors. Its loss, or the loss of its queue, extends the recovery of every facility it supplies, and it carries the full design-basis load.

The second tier is the plant that builds a long-lead component for which spares exist in some specifications, or for which several qualified plants compete, so that its loss lengthens recovery without leaving it unbounded. Most large-power-transformer plants for common transmission voltages fall here.

The third tier is every other plant in the class, including those that build smaller units with shorter lead times. It is in the class, and the class's provisions apply to it in proportion.

## 2.3 Deployment Typologies and Size Range

The class spans several kinds of plant. A transformer plant receives electrical steel and copper, cuts and stacks the core, winds the coils, assembles the active part, dries it in a vapor-phase or vacuum oven, places it in its tank with oil, and tests it in a high-voltage bay before shipping. A turbine and generator plant machines forgings and castings, assembles rotors and casings, balances rotors at speed, and tests the assembled machine where it can. A castings and forgings plant melts and pours steel, forges it under heavy presses, heat-treats and machines it. Many plants serve a world market from a single site; some are owned by companies headquartered abroad.

The range is wide. At one end is a plant that can build only a few of the largest units a year and is the sole source for some specifications; at the other a plant that builds hundreds of smaller units. The class is written to fit both, because the consequence of either is measured in the recovery of the facilities waiting for what it makes.

## 2.4 Consequence Context

The class's consequence is the recovery timeline it imposes on others. The energy series defines RT_HVLLC as the time to restore a facility to full service after the physical destruction of its most difficult-to-replace major component, from destruction to return to operation, and states it for five electric-sector facility classes, from eight to thirty months (*The RT_HVLLC Gap* §1.1, §2). The framework makes the point the class is built on: that timeline is not a property of the damaged asset but is set by manufacturing capacity, so that an adversary who destroys a large power transformer imposes a recovery measured in months, and one who destroys the transformer and the few plants that build transformers imposes a recovery measured in years, or one with no bound at all (framework §1.5).

The record shows the timeline moving without any plant being damaged. The National Infrastructure Advisory Council's report to the President of June 2024 records that utilities which waited months for a transformer as recently as 2020 were waiting far longer by 2024, citing Wood Mackenzie's figures that average lead times rose from about fifty weeks in 2021 to about one hundred and twenty weeks in 2024 and that large units ran from eighty to two hundred and ten weeks, and that transformer prices had risen about eighty percent since the start of the pandemic. The shortage has not ended: Wood Mackenzie projected on 14 August 2025 a supply deficit of about 30 percent for power transformers in 2025, with imports supplying about 80

percent of them, and expects the constraints to persist well into the 2030s (*The Shortage of Transformers in the United States* §3.4). And it shows the timeline under attack. After Russia's campaign against Ukraine's power system began in the autumn of 2022, the national transmission operator reported in January 2023 that it needed fifty-nine large autotransformers, while the country's own largest transformer plant had been reported closed by the war at the start of the full-scale invasion, its status since then open (Section 6.3).

# 3. The Facility-Class Design Envelope

## 3.1 Why Long-Lead Components Are the First Class

Three reasons put this class first. The framework names it the flagship of the sector and the Phase-A instrument, to be developed on the basis of the strategic-reserve work (framework §3.2, §9). It is the class most directly tied to another series: the energy series has already stated RT_HVLLC for its facility classes and has proposed a reserve to bound it (*The RT_HVLLC Gap* §4), and this class supplies the manufacturing side of the same figure. And it is the class in which the framework's organizing insight, that manufacturing capacity is the floor on every other sector's recovery, is most concrete. Precision and Electronics, whose events are better documented, is left to its own class (Section 9.2).

## 3.2 The Two Consequence Chains

Every event in the class reaches the facilities that depend on it by one of two chains.

Chain R, the replacement chain, begins with something that removes or delays the capacity to build a component: the loss of a plant, of its test bay, of its critical material or of its people, or a surge of demand that fills its order book. A facility that loses its component waits longer for the replacement, and every other facility in the queue waits with it. Its consequence is the extension of RT_HVLLC across the population of facilities the capacity serves, measured in months, and it is felt only when a facility needs a unit.

Chain C, the coupling chain, begins with an adversary who destroys an asset and, by the same campaign, the capacity to replace it, or who strikes the installed assets of a sector at a scale that exceeds the replacement capacity the world can supply in the time that matters. The two losses together produce a recovery timeline longer than either alone, and possibly without a bound. It is the case the framework names as the sector's distinctive military-action consequence (framework §1.5, §7), and the war in Ukraine since 2022 is its case.

The two chains are joined by the order book. A plant whose order book is full is already in Chain R for every facility that will need it; an adversary who understands the order book can choose the moment when a strike on the installed assets will find the replacement capacity already committed.

## 3.3 Design Parameters at Class Resolution

Six parameters govern the envelope. Together they are this class's statement of the manufacturing side of RT_HVLLC, which the framework adopts as its native metric (framework §3.6), and Section 10.2 records them for the framework to take up.

The first is the build time: for each component and specification a plant makes, the time from a firm order to a unit tested and ready to ship, with the queue set aside. It is the irreducible part of RT_HVLLC, which adds to it the queue, the delivery and the installation.

The second is the queue time: the time a new order waits before work on it begins, set by the plant's order book and its capacity. It is the part of RT_HVLLC that moves with demand, and in the shortage of 2021 to 2024 it was the part that moved most. Somebody must have the duty to watch it. Every lead-time figure the class could find, the Council's included, came from a consultancy or the trade press, and no public body's continuing measure of transformer lead times was found (*The Shortage of Transformers in the United States* F-1); the class therefore requires that the coordination class hold a refreshed national measure of queue time by first-tier specification and size the reserve coverage against it (*The Shortage of Transformers in the United States* R-1; noted to CorpCo and Project 8).

The third is the qualified source count: the number of plants, anywhere, that can build a given specification and have been qualified by the facility that needs it, with the share of them under domestic or allied control. It sets how much of Chain R a single plant's loss produces, and whether Chain C can make the timeline unbounded. It is stated with its trade exposure: the share of sources and critical materials subject to trade measures, and the effect a change in those measures would have on queue and cost, because imports supply about 80 percent of United States power transformers and tariffs moved cost and queue in peacetime (*The Shortage of Transformers in the United States* F-2, R-2).

The fourth is the surge capacity: the additional units a plant, or the nation's plants together, could build in an emergency, by priority allocation, extra shifts or the conversion of other lines, and the time to reach it. The Defense Production Act gives the government the authority to direct priority and to finance capacity (Section 8.4); the surge capacity is what that authority can actually produce.

The fifth is the reserve coverage: the fraction of the facilities a plant serves for which a compatible spare is held, in the utility's yard, in a pooled reserve or in a federal reserve, so that the facility's recovery does not wait on the plant at all. The framework names the strategic-spares reserve as the measure that decouples RT_HVLLC from the loss of any single plant (framework §3.6), and the energy series' reserve proposal states how it would be held (*The RT_HVLLC Gap* §4). The sixth is the repair capacity: the units a damaged facility can have repaired rather than replaced, the time to repair them, and the plants, crews and parts that repair depends on. Repair is part of the replacement path, because in Ukraine 42 of 94 vital high-voltage transformers were damaged or destroyed between January and April 2023 and more than half of those damaged

were struck again, which blocked their repair (*The Denial of Large-Transformer Replacement in Ukraine* F-1, R-1). A plant states the repair it supports for the units it builds, and under military action repair is credited only where it is protected from repeat strikes (Section 6.3). The parameter is proposed here and recorded for the framework (Section 10.2).

The framework asks for the design basis of a long-lead-component plant to be stated in terms of the RT_HVLLC it imposes on dependent sectors if its capacity is lost (framework §3.6). The class states that figure as the queue time plus the build time plus the delivery and installation time, for the facilities not covered by a reserve or served by repair, after the plant's loss has removed its share of the qualified source count; and it states the class's requirement that, for first-tier components, the reserve coverage be sized so that no facility of the highest consequence tier waits on a plant at all.

This instrument states the parameters and their governing logic. Their quantitative values for a plant or a component await validation against what manufacturers, utilities and the government already hold: order books and delivery records, the energy series' RT_HVLLC values, the spares utilities and the industry program hold, and the industry surveys the energy series' reserve paper calls for (Section 10.1). They are not asserted here as numbers, and the figures quoted are their sources' own.

# 4. The Order Book and the Replacement Path

## 4.1 Status and Use

Every event in this class reaches the facilities that depend on it through the path a replacement unit travels: from the moment a facility needs a unit, through the queue, the build and the test, to delivery and installation, or around all of them by a spare already held. The path is where the class can bound both chains, and this section states what the class requires of it. It is written as the receiving function of the hospital class, DBA-HPH-CD-FC1 (*Care Delivery*), is written at its §4, because it is the plant's interface with the facilities it serves.

## 4.2 The Queue as a Design-Basis Quantity

The plant's order book is usually treated as a commercial matter. In this class it is a design-basis quantity, because it sets the queue time of Section 3.3 for every facility that may need a unit. The class requires that a first-tier plant's design basis state, for each specification it builds, its build time, its current queue time and the queue time at which a facility of the highest consequence tier would wait longer than its own design basis allows; and that it state how an emergency order would be placed ahead of the queue, by whose authority and at whose cost to the orders displaced. The Defense Production Act's priorities system gives the government the authority to direct such placement (Section 8.4). The class requires that its use for a long-lead component be planned before it is needed, because a priority directed in the week of a loss is directed into a queue nobody has examined. The emergency order path includes the buyer and the funds. In Ukraine the first donor-funded autotransformers arrived about nine months after the transmission

operator stated its need (*The Denial of Large-Transformer Replacement in Ukraine* F-2). The class therefore requires that the path name in advance who buys an emergency unit, with what funds and under what procurement rules, and which qualified plants beyond the reach of a campaign it would buy from (*The Denial of Large-Transformer Replacement in Ukraine* R-2; noted to CorpCo).

## 4.3 Qualification and Standardization

A unit built for one site may not fit another, and a facility can receive a replacement only from a plant it has qualified and in a specification it can accept. The fewer the specifications, the more facilities a given unit or spare can serve, and the more plants can build for each. The National Infrastructure Advisory Council recommended standardized transformer designs and interfaces in 2024, and the energy series' reserve paper builds a standardization incentive into its proposed reserve (*The RT_HVLLC Gap* §4.3). The class requires that a first-tier plant's design basis state the specifications it can build and the facilities that could accept each, so that the qualified source count of Section 3.3 is known for each specification and not only for the component.

## 4.4 The Reserve as the Bound on RT_HVLLC

A spare held in advance takes a facility out of the queue altogether. The framework names the strategic-spares reserve as the measure that decouples RT_HVLLC from the loss of any single plant (framework §3.6), and the energy series has proposed a reserve architecture with a priority layer owned by utilities and held in federal custody, a federal buffer layer, and a Defense Production Act surge backstop (*The RT_HVLLC Gap* §4). The industry's own Spare Transformer Equipment Program, administered by the Edison Electric Institute since 2006, is not a pool but a binding agreement: each participant keeps its own spares and must sell them to a participant whose substations are destroyed or disabled by an act of terrorism followed by a presidential declaration of emergency, so that neither a campaign short of that trigger nor a shortage brings it into play. Ukraine, in 2023, planned to hold replacement transformers outside the country, in Poland and Romania, until they were needed, as The Economist reported in October 2023. This class does not restate the reserve architecture, which is the energy series' and the coordination class's. It states what the reserve requires of the plants: that the specifications held in reserve be ones the plants can build again, that the plants' surge capacity be stated against the reserve's replenishment, and that a reserve drawn down in a campaign be refilled from a capacity that the same campaign has not taken.

## 4.5 Test, Transport and Installation

A unit is not available until it has passed its test and reached its site. A plant whose high-voltage test bay is lost cannot ship what it has built; a unit that cannot travel because its route, its rail car or its port is lost cannot be installed; and a unit that arrives without the crew and equipment to install it waits on the dock. The class requires that a first-tier plant's design basis state its dependence on its test bay and the time to restore it, and that the delivery of its units be planned

with the transportation series' conveyance class for the routes and heavy-haul equipment the largest units need (*Conveyances and Fleet*). In Ukraine, the largest transformers were unlikely to fit available cargo aircraft and would most likely go by sea, a voyage of several months (Atlantic Council, UkraineAlert, 13 January 2023).

### 4.6 Allocation When Demand Exceeds Supply

When many facilities need units at once, as in a campaign against a grid or in a surge of new demand, the plants' output must be allocated, and the allocation decides which facilities recover first and which wait. The decision is among the most consequential the class produces, because it sets the RT_HVLLC each waiting facility will bear. The class requires that the criteria for allocating a first-tier plant's output in an emergency be stated in advance, with the authority that applies them, and that they rest on the consequence each waiting facility's outage carries, as the series that owns each facility states it. Where a component within scope proposes the allocation, the allocation is an action placed on its consequence (Section 7.3).

## 5. Design Basis Events: The Envelope

The events below are the class's design-basis envelope, each stated with the event postulated, the consequence threshold, the function the class requires and the duration for which it is required. Sections 5.1, 5.3 and 5.7 are non-adversarial; Sections 5.2 and 5.4 may be either; Sections 5.5 and 5.6 are adversarial. Inclusion is by downstream recovery-consequence severity, not likelihood (framework §4.1).

### 5.1 Loss of a Plant by Fire or Natural Hazard

**Postulated event** fire, flood, wind or earthquake that destroys or disables a first- or second-tier plant's production, drying or test facilities, or the units in it awaiting completion or shipment.

**Consequence threshold** loss of the plant's share of the qualified source count for its specifications until it is restored, and loss of the units in progress, extending the queue time for every facility it serves (Chain R).

**Required function** the plant states its time to restore against the RT_HVLLC its customers can bear; its customers' orders can be moved to other qualified plants, whose capacity to take them is stated; and the reserve coverage of Section 3.3 holds for first-tier specifications.

**Design-basis duration** until the plant is restored, or until its orders are delivered by others.

The fire at the SGB-SMIT Power Matla transformer plant in Pretoria, South Africa, on 7 March 2021, at a site that builds large power transformers and whose principal customer is the national utility, is noted rather than proposed; several transformers were destroyed, and the plant's production could not be moved to the company's other South African plant, which builds smaller units, as reported (Transformer Magazine, 9 March 2021).

## 5.2 Loss of a Critical Material or Upstream Part

**Postulated event** the plant's supply of a critical material or part, grain-oriented electrical steel, copper conductor, insulating material, bushings or heavy forgings, is lost or delayed, through the loss of a supplier, an export restriction or a disruption of transport.

**Consequence threshold** the plant cannot build at its rate, extending the queue time for every facility it serves (Chain R); where the material has one or two sources, the loss reaches every plant that depends on them.

**Required function** the plant states, for each critical material and part, its sources, the share of each under domestic or allied control, and the stock it holds; and the primary-metals class states the resilience of the sources this class depends on.

**Design-basis duration** until supply is restored or an alternative source is qualified.

The National Infrastructure Advisory Council's report of 2024 recorded that grain-oriented electrical steel for transformer cores is hard to source domestically, and recommended coordinated incentives, efficiency standards and trade policy to secure it. The class records the dependence and refers the material's own resilience to *Primary Metals and Strategic Materials* (Section 8.3).

## 5.3 Demand Beyond Capacity

**Postulated event** demand for a component rises beyond the world's capacity to build it, through new load, the replacement of an aging fleet, the build-out of new generation, or the recovery from a widespread event, so that queues lengthen across every plant.

**Consequence threshold** an extension of RT_HVLLC for every facility that needs a unit while the surge lasts, with no plant having been damaged (Chain R); a facility that loses its unit in that period waits the extended time.

**Required function** the queue time is stated and watched as a design-basis quantity (Section 4.2); the reserve coverage of Section 3.3 is sized for the extended RT_HVLLC rather than the normal one; and the surge capacity of the nation's plants, and the authorities that would expand it, are stated.

**Design-basis duration** the duration of the surge, measured from when queue times first exceed the normal lead time to when they return.

The case is the shortage of large and distribution transformers in the United States from 2021, conferred by the framework as DBA-CM-RE-2 (framework §9.1), whose instrument reads it as non-adversarial and as continuing. The National Infrastructure Advisory Council's report to the President of June 2024 records Wood Mackenzie's figures that average lead times rose from about fifty weeks in 2021 to about one hundred and twenty weeks in 2024, with large transformers from eighty to two hundred and ten weeks, and one domestic plant disclosing a five-year wait for new orders; that prices rose about eighty percent since the start of the

pandemic; that domestic plants supply only about twenty percent of large-power-transformer demand; and that the causes included the disruption of supply chains, an aging fleet, electrification and new loads such as data centers, the build-out of renewable generation, labor shortages, and manufacturers' reluctance to add capacity in a cyclical industry. It recommended federal support for domestic capacity, better forecasting, long-term contracts, a virtual strategic reserve with the government as buyer of last resort, standardized designs, a secure supply of electrical steel and a workforce pipeline. The design-basis reading is that the RT_HVLLC the energy series states as a normal-condition figure more than doubled on average without any attack, so that a facility destroyed in that period would have waited far longer than its design basis assumed, and that the surge came from the same forces, new load among them, that make the grid more critical. The condition continues: Wood Mackenzie projected in August 2025 a supply deficit of about 30 percent for power transformers in 2025, with demand up 116 percent since 2019 and imports supplying about 80 percent, and trade reporting of May 2026 gave lead times of up to four years for high-capacity units (*The Shortage of Transformers in the United States* §3.4). Whether the Council's virtual reserve or any federal reserve now exists was not established.

## 5.4 Loss of the Test Bay or the Means of Delivery

**Postulated event** the plant's high-voltage test bay or test stand, or the rail cars, trailers, routes or port facilities by which its largest units move, are lost or unavailable.

**Consequence threshold** completed units cannot be proven or delivered, extending RT_HVLLC for the facilities awaiting them (Chain R).

**Required function** the plant states its dependence on its test facility and the alternatives; and the delivery of first-tier units is planned with the transportation series for the routes and equipment they need (Section 4.5).

**Design-basis duration** until testing or delivery is restored or an alternative is in use.

## 5.5 Cyber Attack Damaging Production Equipment

**Postulated event** an adversary reaches the control systems of a plant's heavy production equipment, a furnace, a press, a vapor-phase oven or a test bay, and causes it to fail or to be damaged, or prevents its safe shutdown.

**Consequence threshold** physical damage to the plant's production capacity, removing its share of the qualified source count until the equipment is repaired or replaced, itself possibly a long-lead item (Chain R); and harm to the plant's people where the equipment is hazardous.

**Required function** the safe shutdown of hazardous production equipment does not depend on control systems reachable from the plant's business network (Section 7.2); the plant states its time to restore each item of heavy equipment, and which of them are themselves long-lead; and access from the business network to the production network is controlled and monitored.

Unexplained control failures are treated as a possible attack: the plant holds criteria at which hazardous equipment is taken to a safe state by independent means when its controls fail without explanation, decided by a named authority, because at the steel mill failures grew in frequency before the furnace could not be brought to a safe state (*The Cyberattack on a German Steel Mill* F-1, R-1). And what the plant knows about its process is protected like the process: the documents and systems that describe its control logic, protective settings and the operating limits of its heavy equipment are identified and protected, and their exposure is treated as a loss of protection (*The Cyberattack on a German Steel Mill* F-2, R-2; noted to *Primary Metals and Strategic Materials*).

**Design-basis duration** until the damaged equipment is restored.

The case is the attack on a German steel mill reported by the Federal Office for Information Security in its annual report on the state of information security in Germany of December 2014, conferred by the framework as DBA-CM-RE-3 (framework §9.1). On the Office's account, the attackers used spear-phishing and social engineering to reach the plant's office network and from it the production network; failures of control components and of whole systems followed; and a blast furnace could not be shut down in a controlled way, which caused massive damage to the plant. The Office described the attacker's technical skills as very advanced, with extensive know-how of the industrial control systems and the production process. It did not give the date of the attack or name the plant or the attacker, and this instrument makes no finding of either. The design-basis reading is that an attack through the business network reached the control of heavy production equipment and defeated its safe shutdown, so that the damage was to the capacity itself; that a plant of this kind feeds the castings and forgings on which the long-lead class depends; and that the repair of damaged heavy equipment can itself be a long-lead problem.

## 5.6 Compromised Units Delivered to the Facility

**Postulated event** a unit leaves the plant carrying a defect, an altered control or monitoring component, or firmware that is not the manufacturer's, so that it fails or can be made to fail after installation.

**Consequence threshold** the loss of the facility the unit serves, after installation and possibly at an adversary's choosing, with the RT_HVLLC of a replacement (Chain R); and loss of confidence in every unit of the same make.

**Required function** the plant controls and records the provenance of the control, protection and monitoring components it fits, and the firmware they run; the factory acceptance test includes them; and the plant can state, for a unit already delivered, what it contains.

**Design-basis duration** the service life of the unit.

The firmware and electronics of a delivered unit are products of the information-technology series' supply-chain class (*Software, Hardware and Product Supply Chain*, DBA-IT-SW-FC3)

and of the precision-and-electronics class; this class requires that the plant be able to state what it fitted.

## 5.7 Loss of the Skilled Workforce

**Postulated event** the plant cannot staff its design, winding, assembly or test functions, through retirement without replacement, competition for skilled labor, illness or evacuation.

**Consequence threshold** a fall in the plant's output and a lengthening of its queue (Chain R).

**Required function** the plant states the staffing on which its build time and surge capacity depend, and the time to train replacements.

**Design-basis duration** the duration of the shortage.

Table 5-1. Facility-class design-basis event categories.

| Event category | Postulated event (class-specific) | Consequence chain | Framework source |
|---|---|---|---|
| Loss of a plant by fire or natural hazard | Production, drying or test facilities destroyed | Chain R | §4.1 candidate (loss of a limited-source plant) |
| Loss of a critical material or upstream part | Electrical steel, conductor, bushings or forgings lost or delayed | Chain R | §4.1 candidate (single-source material); §3.3 |
| Demand beyond capacity | Queues lengthen across every plant | Chain R | developed here; §1.5 |
| Loss of the test bay or the means of delivery | Units cannot be proven or moved | Chain R | §6.2 (heavy-haul boundary); developed here |
| Cyber attack damaging production equipment | Heavy equipment damaged or its safe shutdown defeated | Chain R | §5; developed here |
| Compromised units delivered | Unit carries an altered component or foreign firmware | Chain R (after installation) | §5 (firmware trustworthiness); developed here |
| Loss of the skilled workforce | Design, winding, assembly or test unstaffed | Chain R | developed here |

# 6. Military-Action Extension: DBA-MA for Long-Lead Components

## 6.1 MA Extension Applicability Determination

Every first- and second-tier plant in the class performs the facility-level applicability determination the series makes mandatory, under the two-factor test of geolocation and strategic value, with strategic value driven by irreplaceability and by the breadth of the downstream recovery its loss would deny (framework §7). The framework states a strong sector-level presumption toward applicability for the limited-source long-lead-component capabilities (framework §7). For this class geolocation includes not only where the plant is, but under whose law it operates and where its critical materials come from, because a plant abroad, or one dependent on a material controlled abroad, can be denied by a decision taken there (framework §6.3). The determination is recorded, it is time-varying and re-triggered on the indicators the DBA-MA parent framework defines, and any scope reduction carries a recorded basis (Section 6.9).

## 6.2 The Four Characteristics Applied to the Class

Applied to long-lead components, the four military-action characteristics take specific form, in the canonical statement of FD-DBL (*Design Basis Layers*), at its §4.4. Intentionality: the adversary chooses the installed assets whose replacement is slowest, and the plants and materials that bound their replacement, because the strategic objective is not the event on site but the denial of recovery (framework §7). Persistence: the adversary strikes repeatedly, over a season or longer, so that the installed base is damaged faster than the world can replace it, and so that spares are drawn down before replacements arrive. Precision: the adversary chooses the specifications with the fewest spares and the fewest qualified plants, and the moment when the order books are already full. Denial: the adversary takes the plants, their materials, their test bays or the routes by which their units move, so that the damaged assets cannot be replaced at all within the time that matters.

Under the author's rulings of 10 and 13 October 2026, the military-action tier is not met without denial, and all four characteristics must be established on the record; a characteristic not determined counts as not present (*Design Basis Layers* §6.4). On the author's ruling of 14 October 2026 the characteristics may be read against a campaign on a class's dependencies, the elements precision aims at not needing to belong to the facility under analysis (*Design Basis Layers* §6.4); DBA-CM-RE-1 reads Ukraine that way and finds the tier met, all four characteristics established on the record. DBA-CM-RE-3 does not meet it, precision and denial not being determined and the actor unidentified.

## 6.3 The Asset and Its Replacement Struck Together, as the Governing MA Scenario

The governing military-action scenario is the one the framework names: a campaign that destroys the installed long-lead assets of a sector and, by the same campaign, the capacity to replace them (framework §1.5). It opens both chains at once. The installed transformers, autotransformers and generating units are struck repeatedly; the domestic plants that build them are struck, occupied or left in the line of fire; the routes by which foreign units would arrive are threatened; and the spares are drawn down before the first replacements are built. No credit is taken for domestic capacity, routes or spares that the same campaign can reach.

The extension requires three things. It requires that the reserve coverage of first-tier specifications be sized for a campaign rather than a single event, and held where the campaign cannot reach it. It requires that the qualified source count for each first-tier specification include plants beyond the reach of the campaign, qualified in advance, with an agreed path for placing emergency orders with them. And it requires that the delivery of units from those plants be planned for routes that survive the campaign. And it credits repair only where the repair is protected from repeat strikes (Section 3.3).

The empirical case is Ukraine since February 2022, conferred by the framework as DBA-CM-RE-1 (framework §9.1): the closure of Zaporozhtransformator, the country’s largest transformer plant, by the war, together with the campaign against the country’s power system from the autumn of 2022. The plant, in Zaporizhzhia, close to the front, was forced to close its operations because of the war, as reported by Transformer Magazine on 7 March 2022, which described it as the largest transformer factory in Europe, with an annual capacity of 60,000 MVA. Strikes against the power system, which Ukraine and its partners attribute to Russia, began in the autumn of 2022, and the national transmission operator reported in January 2023 that it needed fifty-nine large autotransformers, as reported by the Atlantic Council. The United Nations Development Programme’s damage assessment, released on 20 June 2023, found that 42 of the 94 vital high-voltage transformers in government-controlled territory had been damaged or destroyed between January and April 2023, and that more than half of those damaged had been struck again, which blocked their repair; that is the denial this scenario postulates. Partners and international agencies delivered transformers, among them two autotransformers delivered by the United Nations Development Programme with Japan in October 2023, and in 2023 Ukraine planned to have one hundred new high-voltage transformers made or bought, half at home and half abroad, the foreign units to be held in Poland and Romania until needed, with few domestic units made so far because of attacks on Ukrainian factories, as The Economist reported in October 2023. The design-basis reading is that a country lost its largest domestic source of the component at the start of the war and then faced a campaign aimed at the installed base of that component, so that its recovery depended on foreign plants, foreign spares and foreign routes, which is the coupling the framework predicts. The plant’s status after March 2022 is open: the government took control of the company under martial law in November 2022, and no source

read states when, or whether, the plant resumed full production (*The Denial of Large-Transformer Replacement in Ukraine* §2.2). The instrument states each party's account as that party's.

## 6.4 Denial by Decision

The second scenario is denial without a strike. A plant abroad may be barred from supplying, a critical material may be placed under export control, or a supplier subject to an adversary's law may be directed to delay or withhold. The framework names foreign ownership as a pre-positioning and denial vector distinct from kinetic and cyber attack (framework §6.3). The class requires a first-tier specification's design basis to state, for each qualified plant and each critical material, the law under which it operates, and to treat a plant or material subject to an adversary's control as absent from the qualified source count in the military-action domain.

## 6.5 The Coordinated Cyber Scenario

The governing cyber scenario is the German steel mill attack timed to a campaign against the installed base. An adversary who has prepared access to the production networks of the few plants that build a component, and to the plants that make their forgings and steel, can damage their heavy equipment at the moment it strikes the installed assets, so that both are lost together. Against it the class relies on the separation of the safe shutdown of heavy equipment from networks an adversary can reach (Section 7.2), on the plant's statement of its time to restore each item of heavy equipment (Section 5.5), and on the reserve and the qualified sources beyond the adversary's reach (Section 6.3).

## 6.6 Cross-Sector Interaction

Under military action the class is tied to every sector its components serve, and the tie runs outward. Its consequence is expressed as the RT_HVLLC it imposes on the energy series' facility classes above all, but also on the water, communications and transportation sectors whose pumps, switching and rolling stock the framework names (framework §6.1). Coming in, the class depends on energy for its energy-intensive processes, on the primary-metals class for its materials, on transportation for the movement of its units, and on the information-technology and precision-and-electronics classes for the controls and firmware it fits (framework §6.2). An adversary need not strike a plant to deny its output; a campaign against the installed base that outruns the plant's capacity, while its materials or routes are denied, is enough.

## 6.7 The Adequacy Result: Bounded Recovery Condition

The test of whether the class's protection is adequate is whether, for each first-tier specification, the nation can reach what this document calls the Bounded Recovery Condition. When the installed assets of that specification are destroyed at the scale the design-basis campaign postulates, and any one plant, any one critical material source and any one route are lost with them, every facility of the highest consequence tier is restored from a spare already held beyond

the campaign's reach, and every other facility within the RT_HVLLC its own design basis can bear, from qualified plants and materials the campaign has not taken, by routes it has not cut. The condition bounds Chain R by reserve coverage, qualified source count and surge capacity, and bounds Chain C by placing the reserve and some of the qualified sources beyond the campaign's reach.

The protective functions on which the plants themselves rely are those that keep their heavy equipment from destroying itself and harming its people: the safe shutdown of furnaces and ovens, the limits of presses and cranes, and the interlocks of the high-voltage test bay. These are to be locally sovereign as FD-LS defines the property where they are protective functions in that sense (*Local Sovereignty* §3.2): a measurement against a setpoint, colocated with the equipment, which no system an attacker could reach can stop, delay or trigger, and whose settings change only through a governed act. The steel-mill attack shows the failure of exactly this property (Section 5.5). Local Sovereignty governs and is not restated here. It describes what protects the plant's equipment when its control systems have failed; it does not decide whether any action needs human authorization (*Local Sovereignty* §4). The class's adequacy rests chiefly not on the plant's protection, however, but on the reserve and the qualified sources, which no single plant can provide.

## 6.8 Adequacy and the Declared Residual

In the non-adversarial domain, the Bounded Recovery Condition is achievable with the Section 4 requirements: a stated and watched queue, standardization that widens the qualified source count, a reserve sized for the extended RT_HVLLC of a surge, and planned delivery. Against a deliberate adversary, adequacy has limits.

Three residuals are declared, in the output form DBA-MA-DiD (*Defense in Depth*) sets at its §3.5. First, a campaign that destroys installed assets faster than the reserve and the reachable plants can replace them leaves facilities waiting; beyond the design campaign the residual is an outage measured by the world's surge capacity, and the response is the allocation of Section 4.6 by consequence and the repair of damaged units where repair is possible and protected from repeat strikes (Section 3.3). Second, a specification for which every qualified plant is within the campaign's reach or an adversary's control has no bound on its RT_HVLLC; the residual is unbounded, and the design response is to qualify sources beyond that reach before the campaign begins, which no plant's design basis can do alone. Third, a unit delivered with an altered component may fail at an adversary's choosing years after installation; the residual is the loss of the facility it serves, and the response is the provenance control of Section 5.6 and the reserve. These residuals bound the class adequacy claim.

## 6.9 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter, such as an RT_HVLLC credited at the normal-condition lead time during a surge, a qualified source counted that is under an adversary's control, a reserve credited that is held where the campaign can reach it, a surge capacity credited

without the materials to build it, or a delivery credited over a route the campaign can cut, must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Sections 9.1 and 10.10 record it for the framework to adopt. A repair credited under military action that is not protected from repeat strikes is such a relaxation (*The Denial of Large-Transformer Replacement in Ukraine* R-1).

# 7. The Dual Design Basis, Stated in Full

## 7.1 The Two Design Bases

The plant carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 7.4.

## 7.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions the plant relies on to keep its heavy equipment and its people safe independently of its networked systems, and the controls on the quality of what it builds: the safe shutdown of furnaces, vapor-phase ovens and heat-treatment, the limits of presses and heavy cranes, the interlocks of the high-voltage test bay, and the factory acceptance test that proves each unit. A furnace that trips to a safe state on its own measurement of temperature, with no networked controller able to inhibit or command it, is a candidate for a locally sovereign function. A function meets *Local Sovereignty* §3.2 only if no system an attacker could reach can stop, delay or trigger it, or change its settings except through a governed act. The steel mill of Section 5.5, whose blast furnace could not be shut down in a controlled way after its control components failed, shows a function that did not.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Controllers that behave predictably within a stated range of inputs, and that have been formally tested in the deployed configuration, are outside the Command Broker obligation for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not below the line; they are outside the obligation only because, and only while, their behavior is bounded by that showing (*The Bright Line* §4.5). The exemption rests on the configuration as installed and tested, and it lapses for a configuration changed outside a governed act (Determination of 8 October 2026, A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out). A component that has not been shown to meet the test is covered by the rules, and each of its actions is judged on its consequence. The framework adds that the firmware-trustworthiness concern the energy series established applies with particular force here, because this is the sector that builds the firmware-bearing components other sectors deploy (framework §5); Section 5.6 states the class's requirement.

## 7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems are entering the class: generative design of windings, cores and rotors; autonomous control of drying, heat-treatment and machining; predictive-maintenance systems that can lock out equipment; and supply-chain optimization that schedules production and allocates output. The framework names these and places the Bright Line for any action of a component within the scope of *Layer Decomposition and Determinism* §4.3 whose incorrect performance could damage a limited-source production capability or misdirect strategic-material allocation at or above the severity threshold (framework §5). These systems are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*); CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The coordination class, CorpCo, is the locus at which the sector's placement and obligation are implemented (framework §3.5). Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

Four cases illustrate the application, and none of them is a placement by component. A system within scope that allocates a plant's output among waiting facilities takes an action with each allocation, placed on the consequence of the delay it imposes on the facilities it does not serve first, which is the RT_HVLLC they will bear: an allocation whose consequence can reach the threshold is above the line and requires the authorization of the qualified human the coordination class names as Command Broker, given on a basis the system cannot alter (Section 4.6). A system within scope that changes a drying or heat-treatment profile takes an action placed on the consequence of the change, which can be the loss of a unit and months of its customer's recovery; a change within limits the plant's process authority has validated may be discharged in advance under FD-BL-D1 (*Mode of Discharge*), and a change to the limits themselves, being a change to the pre-authorized envelope, is excluded from advance discharge by that determination's rule R3. A predictive-maintenance system that locks out a press or an oven takes an action placed on the consequence of the lockout to the plant's output. And a generative-design system whose design is accepted into a unit takes, with the acceptance, one action placed on the consequence of the unit's failure in service.

Any output on which an above-line action is grounded carries the indication-integrity obligation of FD-BL-D2 (*Indication Integrity*): an allocation computed from order-book and outage data that may have been altered is not a basis for authorization until the data can be trusted, and an adversary who can alter the data can choose which facilities wait.

## 7.4 The Interaction Point: Allocation Under a Campaign

The two design bases meet when the plants' output must be allocated while the installed base is under attack. The plant's and the coordination class's systems must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization. That obligation

reaches each such action whose consequence meets the threshold, and the actions most likely to do so are the allocation of a first-tier unit among waiting facilities, a change to the process that makes a first-tier unit, and a lockout of the equipment that makes it; each is placed on its own consequence. The physical design basis keeps the plant's equipment safe; the reserve and the qualified sources bound the recovery; the AI-governance design basis guarantees that no proposing system can decide which facility recovers first, where the consequence requires it, without a qualified human authorization the architecture can enforce.

### 7.5 Scope Boundary

This instrument states the physical design basis in full and names the AI-governance design basis without restating it. The Foundational Definitions own the Bright Line, its scope and its obligation; the Command Broker family owns the Command Broker; the coordination class owns the implementation of placement in the sector and the reserve; and the energy series owns the installed assets and their RT_HVLLC. This instrument applies them to the plants that build long-lead components.

## 8. Cross-Sector Dependencies and Regulatory Context

### 8.1 What Other Sectors Depend On: Replacement Capacity

The class's outward view is the defining one (framework §6.1). The energy series depends on it for the large power transformers, converter transformers, turbines, rotors and generators whose RT_HVLLC it states; the water sector for large pumps and motors; communications and transportation for heavy electrical equipment and rolling stock. The framework's forward elevation runs from the dependent sector's consequence tier back to the plant: a plant that is the sole source of a long-lead component for a facility of the highest tier in another series inherits that tier, and the strategic-spares reserve bounds the obligation (framework §6.1). This class confirms the rule and adds that the elevation attaches to the specification, not only to the component, because a plant may be the sole source of one specification among many it builds.

### 8.2 What the Plant Depends On

The plant depends on energy for its energy-intensive processes; on the primary-metals class for electrical steel, copper and heavy forgings; on transportation for the movement of its oversized units; on the chemical sector for insulating oil and materials; and on the precision-and-electronics class and the information-technology series for the controls, monitoring and firmware it fits (framework §6.2). A compound event that degrades the plant and the logistics that move its output extends every downstream recovery timeline at once (framework §6.2).

### 8.3 The Boundaries with the Partner Classes and Series

Five boundaries are drawn. With the energy series: the installed asset and its RT_HVLLC are that series', and the capacity to replace it is this class's; the reserve architecture is proposed in

that series' strategic-reserve paper and its custody is the coordination class's. With *Primary Metals and Strategic Materials* (DBA-CM-PM-FC2): electrical steel, specialty alloys and large forgings are that class's; the castings-and-forgings plant is shared, and its loss is an event in both. With *Precision and Electronics Manufacturing* (DBA-CM-PE-FC3): the control, protection and monitoring electronics fitted to a unit are made there and fitted here. With the coordination class, CorpCo: supply-chain risk management, Defense Production Act mobilization and the strategic-spares reserve are its members. With the transportation series: the heavy-haul movement of oversized units is shared with *Conveyances and Fleet*.

## 8.4 Regulatory and Institutional Context

The class carries no dedicated federal safety regulator (framework §2.1). The Cybersecurity and Infrastructure Security Agency is the Sector Risk Management Agency. The Defense Production Act gives the government the authority to require priority performance of contracts and orders and to allocate materials and facilities, under the Defense Priorities and Allocations System regulation (15 CFR Part 700), and, under Title III, to finance the expansion of domestic capacity. The industry's Spare Transformer Equipment Program, administered by the Edison Electric Institute, binds its participants to sell spares they hold to a participant whose substations are destroyed by an act of terrorism under a presidential emergency declaration. The reliability standard for the physical security of critical transmission stations (NERC CIP-014) protects some of the installed assets whose replacement this class bounds. Foreign investment in the class's plants is screened by the Committee on Foreign Investment in the United States. The National Infrastructure Advisory Council's report of June 2024 recommended federal support for domestic capacity, including the use of Defense Production Act funding, standardized designs and a virtual strategic reserve.

The regulatory gap is the framework's (framework §2.3). No instrument states the manufacturing capacity the nation must be shown to retain in order to bound its recovery from the loss of a long-lead component; no instrument treats the queue time as a design-basis quantity; and no instrument requires the qualified source count of a first-tier specification to include sources beyond the reach of a campaign. The energy series' reserve proposal addresses the installed side of the gap. This class states the manufacturing side as design-basis requirements for the tiers that carry the load.

# 9. Inheritance Map and Anticipated Derivatives

## 9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally. Facility-class instruments add the manufacturing-function-specific consequence chain, expressed in downstream RT_HVLLC terms, the substitutability analysis, and the recovery-capacity and strategic-reserve architecture (framework §8.1). Table 9-1 is the map for this class. Where the

framework is a skeleton, the element is developed here and recorded for the framework to take up.

Table 9-1. Inheritance map for DBA-CM-LL-FC1.

| Element | Source in the framework | Inheritance mode |
|---|---|---|
| Function-primary placement, LL function class | §1.3, §3.1, §3.2 | Inherited unchanged |
| Substitutability and lead-time sub-index | §3.1, §8.2 | Inherited with adaptation; three tiers stated by specification (Section 2.2) |
| Design-basis principles (downstream recovery consequence, not likelihood) | §4.1 | Inherited unchanged |
| RT_HVLLC as the governing metric | §1.5, §3.6; the energy series | Inherited; its manufacturing side stated as six parameters, repair capacity added at v0.2 (Section 3.3); for the framework to take up |
| Event categories | §4.1 (candidates) | Developed here ahead of the framework; for the framework to take up |
| Bright Line and Command Broker architecture | §5; FD-BL; Command Broker family | Inherited with adaptation: allocation of output placed by the RT_HVLLC it imposes (Section 7.3) |
| The order book and the replacement path | §3.2, §3.6 | Locally defined (Section 4) |
| Bounded Recovery Condition | §7 (skeleton) | Locally defined (Section 6.7) |
| Declared residuals and conditionality check | not yet in the framework | Locally defined; recorded for the framework (Sections 6.8, 6.9) |
| Forward elevation to the plant | §6.1 | Inherited and refined to the specification (Section 8.1) |
| Concentration and foreign ownership | §6.3 | Inherited and applied to the qualified source count (Sections 3.3, 6.4) |

## 9.2 Anticipated Derivatives

Plant-, manufacturer- and component-specific design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability

determination for any scope it reduces. The three reference events this class proposed at v0.1, on the author's decision of 14 October 2026, are conferred by the framework at §9.1: DBA-CM-RE-1, *The Denial of Large-Transformer Replacement in Ukraine*, 2022 to 2023 (Section 6.3); DBA-CM-RE-2, *The Shortage of Transformers in the United States*, 2021 to 2025 (Section 5.3); and DBA-CM-RE-3, *The Cyberattack on a German Steel Mill*, reported December 2014 (Section 5.5), read at the boundary with *Primary Metals and Strategic Materials*. Their instruments were written by Project 11 on 14 October 2026, and this revision takes in their two errata and six recommendations. Two events the plan for closing the seven sectors named for this sector are left to *Precision and Electronics Manufacturing*: the fire at the Renesas Naka semiconductor plant in 2021, and the attack on a Toyota supplier in 2022, which the corpus has already recorded as an impact on automobile manufacturing. The facility-class use case for the governing military-action scenario of Section 6.3 is DBA-CM-UC-1, issued under this revision in the fifth wave of the plan.

# 10. Gaps and Open Items

## 10.1 Quantitative Design Parameters

The six parameters of Section 3.3 are stated without values. Their values for a component and specification are to be validated against manufacturers' order books and delivery records, the energy series' RT_HVLLC values, the spares utilities and the industry program hold, and the industry survey of spare posture the energy series' reserve paper calls for.

## 10.2 The Manufacturing Side of RT_HVLLC

The framework adopts RT_HVLLC as its native metric (framework §3.6). Section 3.3 states its manufacturing side as six parameters, build time, queue time, qualified source count, surge capacity, reserve coverage and, at v0.2 on the finding of DBA-CM-RE-1, repair capacity, and states the RT_HVLLC a plant's loss imposes as the queue time plus the build time plus delivery and installation, for facilities without a spare, after the loss has removed the plant's share of the source count. Whether the framework adopts the six, and whether the energy series should state its RT_HVLLC values as normal-condition figures with a surge figure beside them, are for the framework and for Project 8.

## 10.3 The Specification as the Unit of Analysis

The class states its tiers, its source counts and its elevation by specification rather than by component (Sections 2.2, 3.3, 8.1). This is more demanding than the framework's statement by component, and it depends on data, the specifications installed and the plants qualified for each, that the industry holds and has not published together. The industry survey the energy series' reserve paper calls for is the natural means.

## 10.4 A Note to the Energy Series on the Spare Transformer Program

The energy series' strategic-reserve paper describes the Spare Transformer Equipment Program as a Department of Energy program (*The RT_HVLLC Gap*, Executive Summary and §3). The program is the industry's, administered by the Edison Electric Institute, and it is a binding agreement among participants to sell the spares each holds after a terrorist attack under a presidential emergency declaration, not a federal or pooled inventory. This class cites the program as the industry's. The point is for Project 8 to confirm and, if it agrees, to correct at its paper's next revision.

## 10.5 Local Sovereignty in the DBA-MA Sense

Section 6.7 relies on locally sovereign protective functions for the plant's heavy equipment, as FD-LS defines the property. Whether the DBA-MA parent framework's local-sovereignty principle is the same thing FD-LS defines remains the open question the other classes record.

## 10.6 The Precision and Electronics Class

On the author's decision of 14 October 2026, this class is the sector's first, and Precision and Electronics Manufacturing is left to its own class with the plan's semiconductor and automotive events. The firmware and electronics fitted to long-lead units (Section 5.6) are the point at which the two classes meet, and the precision-and-electronics class should state, when written, how it bounds them.

## 10.7 The Reference Events

The three reference events this class proposed are conferred by the framework at §9.1 as DBA-CM-RE-1 to DBA-CM-RE-3, and their instruments were written by Project 11 on 14 October 2026. DBA-CM-RE-1 bounds the Ukrainian condition from 24 February 2022, with its second phase from 10 October 2022, to October 2023, finds the military-action tier met with all four characteristics established, proposes Energy as an impacted sector, and leaves the record open; DBA-CM-RE-2 records the shortage as non-adversarial and continuing, bounded at Wood Mackenzie's release of 14 August 2025, with Energy an impacted sector; and DBA-CM-RE-3 records the steel-mill attack as not meeting the threshold, the plant and attacker unnamed and the date not given. This revision takes in the errata at Sections 1.3, 2.4, 5.3 and 6.3 and the recommendations at Sections 3.3, 4.2, 5.5, 6.3, 6.8 and 6.9. The use case for the governing scenario of Section 6.3 is DBA-CM-UC-1.

## 10.8 The Register Row

The DBA Facility Class Register now records this class at v0.1 as an issued working draft under the framework at v0.3; Project 0 entered the row on 14 October 2026.

## 10.9 The Framework's Date

The framework's cover reads August 2026 at v0.3, while its revision history records v0.3 as made in October 2026. The cover date is the framework's to correct at its next revision, which the conferral will be.

## 10.10 What This Class Notes to Other Instruments

Several elements of this instrument belong to other instruments, and this instrument records them as noted without stating them for those instruments. To the framework: the six parameters of the manufacturing side of RT_HVLLC (Section 3.3, Section 10.2), the event categories of Section 5, the three tiers by specification (Section 2.2), the declared residuals of Section 6.8, the conditionality principle of Section 6.9, and the cover date of Section 10.9; the conferral section it also asked for is now framework §9.1. To the energy series, in Project 8: that RT_HVLLC moves with the queue and that average lead times more than doubled in the shortage, so that a surge figure should stand beside each normal-condition value (Section 10.2), the note on the spare transformer program (Section 10.4), and, from the instruments, *The Denial of Large-Transformer Replacement in Ukraine* R-1, that repair capacity is part of the replacement path and that the installed units under repair need protecting from repeat strikes, and *The Shortage of Transformers in the United States* R-1, that the national queue be measured. To CorpCo, when written: the buyer and funds of the emergency path (*The Denial of Large-Transformer Replacement in Ukraine* R-2) and the national measure of queue time (*The Shortage of Transformers in the United States* R-1). To *Primary Metals and Strategic Materials* and *Precision and Electronics Manufacturing*, when written: the dependences and boundaries of Section 8.3, the events left to the latter (Section 9.2), and to the former *The Cyberattack on a German Steel Mill* R-2, that what a plant knows about its process is protected like the process. To the transportation series: that the delivery of first-tier units is part of their recovery timeline (Section 4.5). The register row is entered (Section 10.8).

# References

## DBA-CM Series and Parent Framework

*Critical Manufacturing Sector Framework* (DBA-CM_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

DBA-MA Parent Framework. Eclectic Technologies. Source of the design-basis-under-military-action methodology. FD-DBL (*Design Basis Layers*), held for the edition following the October 2026 Edition, is the canonical statement of the four military-action characteristics (§4.4, §6.4).

*Defense in Depth* (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual output form (§3.5).

*The RT_HVLLC Gap: Why the United States Needs a Strategic Reserve for Electric Sector Long-Lead Components*, the energy series' strategic-reserve paper. Eclectic Technologies, 2026. Held

by Project 8. Source of the RT_HVLLC values and the reserve architecture cited in this instrument.

*The Denial of Large-Transformer Replacement in Ukraine* (DBA-CM-RE-1), *The Shortage of Transformers in the United States* (DBA-CM-RE-2) and *The Cyberattack on a German Steel Mill* (DBA-CM-RE-3). Eclectic Technologies, 2026. Held by Project 11; conferred at Critical Manufacturing Sector Framework §9.1 and cited by short title and section, finding or recommendation number.

*Conveyances and Fleet* (DBA-TRAN-CV-FC3), *Care Delivery* (DBA-HPH-CD-FC1), *Chemical Production and Synthesis* (DBA-CHEM-PR-FC1) and *Software, Hardware and Product Supply Chain* (DBA-IT-SW-FC3). Eclectic Technologies, 2026. The heavy-haul boundary, the structural model, and the supply chain of fitted firmware.

*Facility-Class Types Are Physical, Coordination and Logical*, author's determination of 30 September 2026, and *A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out*, author's determination of 8 October 2026, both recorded by Project 0. *Closing the Seven Sectors*, working plan of 13 October 2026, Project 9.

## Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

## Regulatory, Standards and Guidance

Defense Production Act of 1950, as amended, Titles I and III; Defense Priorities and Allocations System, 15 CFR Part 700. North American Electric Reliability Corporation. Reliability Standard CIP-014, Physical Security. Edison Electric Institute. *Spare Equipment and Grid Resilience Programs* and its emergency-response programs material, describing the Spare Transformer Equipment Program.

National Infrastructure Advisory Council. *Addressing the Critical Shortage of Power Transformers to Ensure Reliability of the U.S. Grid*, June 2024.

## Events and Investigations

Transformer Magazine. *Largest transformer factory in Europe closed due to war*, 7 March 2022; *Fire at SGB-SMIT Power Matla transformer factory*, 9 March 2021. Atlantic Council, UkraineAlert. *Ukraine struggles to repair power grid as Russian airstrikes continue*, January 2023. *The Economist*, reporting of Ukraine's plan to order one hundred transformers, 2 October 2023. United Nations Development Programme. Press release on the delivery of autotransformers to Ukraine with Japan. Reuters, report of 7 November 2022 on the government's taking control of the company. United Nations Development Programme, energy

damage assessment, released 20 June 2023. Wood Mackenzie, analysis of 2 April 2024 and release of 14 August 2025, as read in DBA-CM-RE-2.

Bundesamt für Sicherheit in der Informationstechnik (Federal Office for Information Security). *Die Lage der IT-Sicherheit in Deutschland 2014*, December 2014, and contemporary reporting of its account of the steel-mill attack.

# Revision History

| Version | Date | Author | Description |
|---|---|---|---|
| 0.1 — DRAFT | October 2026 | Tim Roxey | Initial issue. First facility-class instrument in the DBA-CM series and the framework's Phase-A instrument, typed physical under the author's determination of 30 September 2026, in the ten-section structure of DBA-HPH-CD-FC1 and DBA-CHEM-PR-FC1. Written under the plan Closing the Seven Sectors, wave 3, on the author's decisions of 14 October 2026: Long-Lead Critical Components as the sector's first class, ahead of Precision and Electronics; three reference events proposed, the denial of transformer replacement in Ukraine (from 2022), the United States transformer shortage (from 2021) and the German steel-mill attack (2014), with Renesas and Kojima left to Precision and Electronics. States the manufacturing side of the energy series' RT_HVLLC as five parameters, the order book and the replacement path as the governing pathway, and the Bounded Recovery Condition; records for Project 8 that the Spare Transformer Equipment Program is the industry's. Checked before issue by an independent review agent, not a person, against primary sources and for internal consistency; its substantive findings were taken in, among them the Council's report correctly attributed and noted as published in draft, and the Spare Transformer Equipment Program described as the binding agreement with a terrorism trigger that it is. |
| 0.2 — DRAFT | October 2026 | Tim Roxey | Conforms to the framework at v0.4 and takes in the return of Project 11's Critical Manufacturing batch and the author's ruling of 14 October 2026 on reading the characteristics against a class's dependencies. The three reference events proposed at v0.1 are stated as conferred, as DBA-CM-RE-1 (Ukraine), DBA-CM-RE-2 (the United States shortage) and DBA-CM-RE-3 (the German steel mill), at Sections 1.3, 5.3, 5.5, 6.2, 6.3, 9.2 and 10.7. Erratum 1 states the plant's status after March 2022 as open, with the government's taking control in November 2022, and adds the United Nations Development Programme's damage assessment (Sections 1.3, 2.4, 6.3). Erratum 2 states the shortage as continuing and attributes the lead-time figures to Wood Mackenzie (Sections 2.4, 5.3). The steel-mill refinements add the attacker's know-how and the date not given (Section 5.5). The six recommendations enter at Sections 3.3 (repair capacity as a sixth parameter; the |

| Version | Date | Author | Description |
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| | | | national queue measured; the source count with its trade exposure), 4.2 (the buyer and the funds), 5.5 (unexplained control failures; what the plant knows), 6.3, 6.8 and 6.9 (repair credited only where protected). Section 10.10 carries the notings to Project 8, CorpCo and Primary Metals and Strategic Materials. The cover lineage stays as issued. The parameter set grows from five to six; no change to the class's scope, chains, event categories or residuals; the v0.1 entry is left as a record. |